

Investigating the Correlation Between Presence and Reaction Time in Mixed Reality

Anonymous submission

Abstract—Measuring *presence* is critical to improving user involvement and performance in Mixed Reality (MR). However, current methods utilize questionnaires that do not capture time-varying responses and are prone to user bias. Inspired by the existing literature on the relationship between *presence* and human performance, we argue that there is a potential to leverage existing user interactions in the virtual scene as a measure of *presence*. We explore the user *reaction time* as one such quantity that can be easily measured using the systemic tools available in modern MR devices. We conducted an exploratory study (N=40) and found a significant correlation between *presence* scores and *reaction times* with a correlation coefficient of -0.65. While our study points to an inverse relationship between *presence* and *reaction time*, we also note that the slope of the negative correlation may depend on the scenario under consideration and needs further investigation.

Index Terms—Mixed reality, Presence

1 INTRODUCTION

AFTER the popularity of Virtual Reality (VR) and Augmented Reality (AR), Mixed Reality (MR) is gaining importance in science, education, training, and entertainment, affording new ways of interaction and engagement with the real and virtual worlds. Recent advances in tools and technology for MR have made it possible for a user to feel “present” in the mixed reality environment. The ability of an individual to behave in a virtual environment as they would in the real world is known as *presence*. *Presence* is typically described as the subjective experience of being in a simulated place or environment and the user’s readiness to respond to virtually generated sensory data as if they were real [1], [2], [3]. This includes being able to interact with and respond to virtually generated sensory data in a natural and appropriate way. As if in the real world, an individual should be able to bend down, grab an object on the floor in a virtual environment, feel its weight, and lift it if they desire to do so. This is achieved through the sense of one’s own body movement and position, that is matched to the sensory data presented in the virtual environment. High *presence* does not necessarily require high fidelity to physical reality, but rather that individuals are able to behave as if the sensory data they are experiencing is real. This approach to measuring *presence* allows for observation and evaluation of an individual’s behavior in both real and virtual environments. A high *presence* is desired in any simulated virtual environment, as it allows the user to experience an immersive, realistic, and involved experience.

There are many studies that investigate the notion of *presence* and describe the factors that contribute to the sense of *presence*. According to Witmer & Singer [4], Slater & Steed [5] and others [1], [2], [3], [6], [7], there are two main aspects of *presence*: *place illusion* and *plausibility illusion*. *Place illusion* refers to the *sense of being there*. In an MR environment, this corresponds to the extent to which the virtual content appears to be indistinguishable from the real world. The second factor is *plausibility illusion*, which refers to the sensation that the observed events depicted

in a virtual environment actually occur. When both *place illusion* and *plausibility illusion* occur, users will feel involved in the environment and will respond realistically to the environment, leading users to experience greater engagement in a MR environment. Both immersion and participation are necessary to experience *presence* [4].

A prerequisite to improving *presence* of a user is the ability to measure and quantify *presence*. While the conventional measures of *presence* have been defined for virtual environments that surround and isolate a user from the real world [8]. We are measuring *presence* as the subjective experience that a particular object exists in a user’s environment, even when that object does not [9]. Due to *presence* subjective nature, the most popular method to measure *presence* is the use of subjective questionnaires [4], [10], [11], [12], [13]. Questionnaires ask users to self-report their sense of *presence* answering questions that attempt to assess *presence*, usually after the user has left the virtual environment. The benefit of subjective questionnaires is that they can be quickly administered, graded, and interpreted without affecting the user experience. However, the questionnaire cannot measure the time-varying qualities of *presence* and can produce unstable, inconsistent, and irreproducible responses due to the prior experience of the participants [14]. The well-known shortcomings of *presence* questionnaires have led researchers to explore alternative approaches to assessing *presence*, including behavioral responses such as postural response [15], hand and eye response [16], startle response [17], which measure participant responses that are produced automatically, without conscious thought, thus avoiding user bias. However, they are susceptible to experimenter bias and are highly sensitive to environmental factors and content [18], [19]. There are also various physiological responses that can be measured to assess *presence*, such as a change in heart rate [20], [21], skin conductance [22], and body temperature [23]. However, physiological measures can also be noisy and unreliable, especially under non-stationary conditions, and may not capture differences in

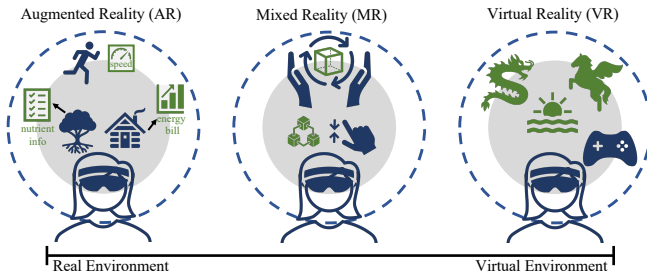


Fig. 1: Physical and virtual reality continuum. Virtual objects are colored green and real-world objects are colored blue.

presence in situations of low emotional valence [22].

Given the state-of-the-art, there is a need for an approach to quantifying *presence* that is objective, discrete, not consciously affected by the participant and/or experimenter, and could be used at runtime without interfering with the virtual experience. It should take advantage of existing interactions (and underline quantities) that are happening in the virtual scene and measure *presence* through these interactions rather than making additional external interventions. One of such underlying quantities is *reaction time*. *Reaction time* or response time refers to the time taken between when humans perceive something and when humans respond to it. *Reaction time* is dictated by the cognitive ability to detect, process and respond to a stimulus [24], [25], and can be easily measured using the systemic tools available in modern MR devices such as Microsoft HoloLens 2 [26].

Our work investigates a fundamental question in mixed reality: Would an individual experiencing more *presence* systematically show faster *reaction times*? If the answer is yes, we could use a systemic metric such as *reaction time* to quantify *presence* in a non-intrusive, objective, and unbiased manner. There is a large body of work investigating the relationship between *presence* and human performance [27], [28], [29], [30], [31], [32], [33]. Natalia et al. showed a negative correlation between task completion time and *presence* when the sense of *presence* is altered by multisensory feedback [31]. Matteo et al. show a negative correlation between performance and *presence* when the sense of *presence* is changed by varying the perceptual load [32]. Maneuvrier et al. showed that *presence* promoted spatial cognition performance and that the *presence*-performance relationship was not mediated by other human factors [28]. Furthermore, human performance is often used as an argument for the good predictive validity of questionnaires [13].

To understand the relationship between *presence* and *reaction time*, we conducted a study in which we change *presence* of users by manipulating *place illusion* and *plausibility illusion* when interacting with an MR environment. We designed two sets of experiments. In one set, we only manipulated the appearance of the virtual object and, in the other set, we manipulated a non-task-relevant behavior of the virtual object. All other aspects of the experiments, such as the interaction mechanism, frequency of interactions, and physical environments, were kept the same. We systematically measure the *reaction time* of users in response to a visual stimulus. Our post-experience questionnaires show a significant change in the *presence* between experiments. Similarly, we see a significant change in user *reaction time* as the sense of *presence* changes across experiments. Our

analysis shows a correlation between *presence* and *reaction time*.

In our attempt to understand the *presence* and *reaction time* relationship, we make the following contributions.

Contribution 1: We propose the use of *reaction time* as a measure of *presence*. We also develop a non-intrusive, systemic approach to measuring the user *reaction time* that relies on existing interactions in MR environments.

Contribution 2: We devise experiments that alter the sense of *presence* by manipulating *place illusion* and *plausibility-illusion*. In designing experiments, we control for other factors that are known to impact user performance. While we use the experiments to demonstrate change in *presence* as a byproduct, we also demonstrate that *presence* questionnaires typically used in fully virtual environments can be used in mixed reality environments as well.

Contribution 3: We conduct an exploratory lab study ($N = 40$) with our setup, demonstrating a negative correlation between the sense of *presence* and the user *reaction time* when responding to a visual stimulus.

2 BACKGROUND AND RELATED WORK

In this section, we define the relevant terminology, discuss the existing work on the concept of *presence* in mixed reality, and present existing methods on quantifying *presence*.

2.1 Terminology

There are different forms of *reality* depending on how much of the physical world is part of the user's experience and how the user interacts with the virtual objects in the scene. Figure 1 shows the physical-virtual reality continuum. We define the different reality environment next.

- **Virtual Reality (VR)** completely replaces a user's view, immersing them within a computer-generated virtual environment. It is a fully enclosed authentic experience with no sense of real world.
- **Augmented Reality (AR)** enhances the user's view of the real world by overlaying computer-generated information. Real world remains central to the experience with limited or no interaction with the virtual environment.
- **Mixed Reality (MR)** is a spatial alignment between the real and virtual world. Real and virtual worlds are intertwined and the ability for users to be able to interact and manipulate both the real (physical) and virtual environment.

In summary, VR environments typically replace the real world with a virtual one through immersion of the user's senses. AR and MR supplement the real world with digital information, affording users real-time interaction with co-existing real and virtual elements.

2.2 Factors Affecting Presence

Presence is a phenomenon of awareness based on the interaction between sensory stimulation, environmental factors that encourage involvement and allow immersion, and internal tendencies to become involved and interact with virtual objects [4]. Sheridan [34] laid the foundation for determining the underlying factors of *presence*, such as sensory information, sensor control, and motor control.

Slater and Wilbur [35], [36], expanded on Sheridan's work to determine major factors affecting user presence. According to them there are two main factors that contribute the most in user presence. (1) *Place illusion* (appearance of virtual object) can be affected by the realism of the virtual content, consistency of the view between headset and direct view from the users' eyes (including displacement or latency issues), lack of haptic feedback, and being overly aware of the headset. (2) *Plausibility illusion* (behavior of virtual object), can be affected by scene elements not obeying the laws of physics, cause-effect relationships not coupled as expected, actions that do not have the expected outcome, and events in the virtual environment not conforming to familiar expectations. It is argued that when both *place illusion* and *plausibility illusion* occur, users will feel involved and respond realistically to the environment, which will lead users to experience greater engagement in a virtual environment [4].

2.3 Presence Measures

Previous research highlights the importance of presence as an outcome of virtual environments (AR/VR/MR) [1], [2], [4], [34], [37] and multiple subjective and objective approaches have been developed. Most commonly accepted method of evaluating *presence* is the self-report questionnaire [38]. The presence questionnaires are used after participants engage in an AR/VR/MR environment (a post-experience questionnaire). Over time, previous work developed, refined, and validated questionnaires to measure presence. Earliest approaches to measuring presence by varying the update rate of a computer-generated simulation on the sense of presence in a virtual environment [39]. They measured the fidelity of the interaction using two questionnaires.

The Witmer & Singer Presence Questionnaire (PQ) is one of the most widely used [4] to test involvement/control, natural, and interface quality. Other popular post-experience questionnaires include the Igroup Presence Questionnaire (IPQ) [10], [11] and the Slater-Usch-Steed Questionnaire (SUS) [5]. The PQ questions are based on the following factors: ability to control the environment and "naturalness" of control over the environment; coherence, consistency of information arriving from different senses, and distractions a participant may experience in a virtual environment; environment realism and meaningfulness; disorientation when returning to the real world. The SUS and IPQ questions are based on three factors: the sense of physically being in the virtual environment, the extent to which the virtual environment feels real, and the extent to which the participant feels involved in the environment. More questionnaires were developed that more or less measured aspects of realism relating to the virtual scene such as ITC-Sense of Presence Inventory (ITC-SOPI) [40], Kim and Biocca questionnaire [41], Object presence questionnaire [8], reality judgment and presence questionnaire [42], Swedish viewer-user presence questionnaire (SVUP) [43], and others [6], [13], [27], [44], [45], [46], [47], [48], [49], [50]. However, the questionnaires cannot measure the time-varying qualities of the presence, can produce unstable, inconsistent, and unrepeatable responses, and are prone to user bias. To mitigate

the temporal information issue, Schwind et al. [51] used integrated questionnaires to measure presence. Integrating questionnaires directly into the virtual experience has been explored to assess virtual experience as the user is going through it [51], [52], [53]. All three studies did not have a clear baseline.

Other subjective measures are continuous assessments [6], [54], psychophysical measures like cross-modality matching; free-modulus magnitude estimation; and paired comparisons. Subjective qualitative method include auto-confrontation [55], focus group exploration [56], interaction analysis [57]; free format self-reports [58]; and repertory grid technique [59]. There are also several subjective corroborative measures to evaluate presence indirectly such as Break In Presence (BIP) [60]; duration estimations by users [61]; attention awareness [62], [63]; and simulator sickness questionnaire [12]. Most of these methods asked participants to complete questionnaires that often add to the phenomenon of break-in presence [60] and are prone to participants' bias.

Objective measures are mostly captured by evaluating the behavioral and physiological responses of the users and are often used as corroborative measures. Popular physiological measures are cardiovascular measures [21]; skin measures [22]; ocular measures [64]; and facial electromyography [65]. Presence is also known to be measured through neural correlates like Electroencephalogram (EEG) and Functional Magnetic Resonance Imaging (fMRI) [66]. Some commonly used behavioral measures are based on assessing facial expression [19]; postural [15]; startle [17], reflex and social responses; and pointing conflicting cues [18].

Behavioral and physiological measures offer relief from user bias but can also often be noisy and unreliable [22].

2.4 Presence and Performance

Many studies have discussed the association of presence in the context of a task performance in virtual environment [5], [67] only a few of them are described here, Mel et al. conducted an experiment to assess the influence of presence on performance, while the participants learned to play three-dimensional chess [68]. They noted that presence refers to behavioral and psychological responses of people. Similarly Barfield & Weghorst work [27] suggested that task performance measures can be used as objective corroborative indicators of presence. Few such methods are task completion time and error rate [69]; number of actions [68]; and secondary task performance [70]. Though it is generally assumed that higher levels of presence are associated with better task performance [71], the exact causal link between presence and task performance is unclear.

Slater et al. [68] explored the relationship between presence and performance. While keeping other factors such as relevant background knowledge and users' ability same, results suggested that presence increased in terms of richness of the portrayed virtual environment did indeed improve task performance. The study also found that reported presence was higher for egocentric compared to exocentric immersion, but that casual relationship was not between presence itself was not associated with task performance [36]. It is also noted that motor behaviour is strongly influenced by perceptual uncertainty and the expected con-

sequences of actions [72] which can affect the characteristics of the user, such as ability and motivation, that will influence task performance [73]. IJsselstein [70] noted that it is reasonable to assume that several characteristics of a virtual environment will similarly influence presence and task performance. They further expanded that performance on a secondary task can serve as a measure for the amount of effort and attention allocated to the primary task. The more effort is dedicated to the primary task, the more performance on the secondary task will decrease. A similar argument can be made in the case of presence: if more attention is allocated to the mediated virtual environment, performance on a secondary task will decrease. Szczurowski and Smith made the case that it is reasonable to assume that the nature of presence is subconscious [74]. Their argument is that if the presence exists outside of the subjective feeling domain, it's unlikely to be a conscious process. No one must remind themselves about staying present in the real world. It's also unlikely that it is possible to force yourself into feeling present in a virtual environment. So IJsselstein, Szczurowski, Smith and others [25], [74], [75], [76] concluded that the reaction times or error rates can be used as task performance measures used for presence evaluation.

Therefore, we make the case for measuring subjective feelings of presence with an approach that is objective, reproducible, and produces a stable response without creating interference in the virtual scene. Despite being aware of problems associated subjective measurement method, we are deciding to use questionnaires as our baseline for our current study, which allowed us to test our hypotheses about relations between different constituents of *presence* and users' *reaction time*.

2.5 Presence in Less-immersive AR/MR Environments

Presence governs aspects of autonomic responses and behaviour of a user in a virtual environment, whereas immersion refers to a quantifiable description of a technology [77]. Wilbur and Slater argue that degree of immersion can be objectively assessed as the characteristics of a technology [36], and has dimensions such as the extent to which a display system can deliver an inclusive, extensive, surrounding and vivid illusion of virtual environment to a user [78]. AR/MR elicits a different sense of presence: "It is here" presence [79]. Although from a technical perspective AR/VR/MR are very different, they share a common feature. It is the curated environment either real (in case of AR and MR) or virtual, that the virtual objects are placed in. Therefore, the common approach to measure presence in the range of virtual environments from least immersive environment (AR) to fully immersive environment (VR), questions on the experience whether one has the sense of being there in the virtual environment or interacting with virtual objects placed in the real environment. Although there have been attempts to develop presence methods [8], [80], [81] and measurement tools [9], [82] exclusively for AR/MR, but there have been criticisms that all of these tool measuring factors influencing presence rather than subjective feeling of presence [83].

While conventionally presence has been defined for virtual environments that surround and isolate a user from

the real world [8]. Slater et al [38] conducted a study on VR questionnaires (PQ and SUS) and concluded that the questionnaires that are developed for VR can still be useful when all users experience the same type of environment even if the environment is not fully immersive (AR/MR). They further concluded that the utility of questionnaires may be doubtful for the comparison of experiences across environments - such as immersive virtual compared to real, or desktop compared to immersive virtual, or real environment with virtual objects to fully virtual environment. Presence questionnaires are often utilized in research to explore the subjective experience of presence rather than the link between perceived presence and aspects of technology; therefore, they can be employed anywhere on the virtuality continuum in technological or real-world context. To this end in this study we are referring to *presence* as the subjective experience that a particular object exists in a user's environment, even when that object does not [9]. This definition is more appropriate for assessing non-immersive displays such as AR/MR headsets [81]. For the rest of the paper we refer to this definition of *presence*.

2.6 Taxonomy of a virtual scene

As described by Wilbur and Slater [36], the place illusion (appearance realism) and the plausibility illusion (behavioral realism) are the main aspect of any virtual experience in any alternative reality medium (AR/VR/MR).

- **Virtual scene** represents the semantics of the virtual environment in three dimensions (3D) placed on top of the real environment.
- **Event** is an action or occurrence recognized by the computer system. For example, when a virtual object is added or removed from the virtual scene, this can be dubbed as an event.
- **Task** refers to an activity that is observable and has a start and an end point. In MR, tasks will be aligned with the start or end of an event, depending on the semantics of the virtual scene.
- **Interaction** is defined as the act of performing a physical action to perform the task in a virtual scene.
- **Cue** is defined as the signal (visual or auditory) that is sent to the participant to initiate a task.
- **Feedback** is defined as the visual or auditory confirmation sent to the participant that the task is completed.

To create a virtual scene that has context, immersion, and interaction, we need to craft our experiments so that participants feel engaged in the virtual world. However, the scene should neither be too complex that the participants' cognitive load is consumed in understanding the scene, nor should the scene be too simple that the participant feels disengaged [70], [77]. For example, we do not want a scene with warcraft heavy-load games, nor do we want a simple box that has no semantic value to the participant. We need to create a balance between a semantically too complex and a simple virtual scene. The same applies to the events, tasks, cues, interaction, and feedback that arise from this virtual scene.

3 APPROACH

To examine the validity of *reaction time* as a measure of *presence*, we investigate *correlation between presence and reaction*

time in mixed reality (MR). For any measure to be useful, Insko et al. [6] suggest the following criteria (C).

- **C1: Sensitivity** - a measure should be able to distinguish between various feelings of *presence*. To test this, we need to create varying feelings of *presence*.
- **C2: Reliability** - a measure should produce repeatable results across trials and across participants.
- **C3: Validity** - a measure should be able to quantify subjective *presence*, or at least correlate well with established subjective *presence* measures. We need to establish a baseline from existing questionnaire based measures and demonstrate correlation with those measures.
- **C4: Objectivity** - a measure should be free from participant and experimenter bias. The tasks and scenarios that we create should not offer an advantage to a certain group of users over others. Tasks should also not be deliberately designed to push the results in the favor of the experimenters' hypotheses.

To meet the above mentioned criteria, we face the following design challenges (DC).

- **DC1:** How can we induce different feelings of *presence*? (C1: sensitivity)
- **DC2:** How do we *minimize external variables* while varying feelings of *presence*? (C2: reliability)
- **DC3:** How do we *establish a baseline* and what that baseline should be? (C3: validity)
- **DC4:** What should the *interaction mechanism* be? (C4: objectivity)

3.1 DC1 - Varying presence

For measuring varying feelings of presence, we need an empirical setup that can measure presence in real-time while depicting a practical scenario. But first, we need to understand what the main aspects of presence are and what dimensions could be measured in those distinct but overlapping aspects. We describe two main aspects (place illusion and plausibility illusion) of presence in Section 2. Since we are using Mixed Reality (MR) as our medium of experiment, place illusion and plausibility illusion would need to be refined. In MR measuring the place illusion could mean to what extent the virtual object is appearing to be indistinguishable from reality. The sense of plausibility can be described when users select a dominant space as the reference frame. Then virtual objects in real space or real objects in a virtual space would be perceived as plausible if the object behaves coherently to the dominantly perceived space as noted in place illusion. For example, gravitational forces are applied horizontally rather than vertically.

In summary, place illusion mainly refers to the elements related to *appearance* of the environment, whereas plausibility illusion refers to the elements related to *behavior* of the objects in the environment. Therefore, we argue that to vary the feelings of presence, we could manipulate *appearance* and/or manipulate the *behavior* of the object in the scene.

3.2 DC2 - Controlling confounding variables

There are a variety of options for manipulating the appearance and behavior of objects in the virtual environment. To satisfy the reliability criteria, we need to carefully pick scenes and manipulations. We have identified two constraints that will help limit the introduction of external

variables and maintain symmetry across experiments and users.

- Constraint 1: Manipulations should not affect (increase or decrease) the overall complexity of the scene, including tasks, events, and interactions. (**simplicity** in scene)
- Constraint 2: Manipulations should also be free from additional confounding variables. Confounding variables are the extra variables that affect the actual relationship between the variables under study [84]. In our case, these two output variables are *presence* and *reaction time*. (**symmetry** in the scene and users)

To satisfy the simplicity constraint, we will avoid unnecessary details in the scene (extra buttons, features) and keep experimental conditions the same across all trials and users. These experimental conditions are the placement and duration of the visual stimulus, time and appearance of the cue, time and appearance of the feedback, number of tasks, duration of the experiment, and type of interaction. We also argue that to keep the scene simple, we should pick a scene that is familiar to most college students (intended participants pool) and have only one virtual object in the scene at a time. However, we want to keep the participant engaged throughout the experiment, and we need a scene with some familiar semantics. Similarly, the manipulations that will vary the presence should be subtle and change only one entity of the scene to isolate the effect on our output variables. In [85] authors have proposed manipulating the appearances of the scene by manipulating the visual fidelity. We can measure the varied presence by making the scene appear *realistic* in one scenario and manipulate the same scene by making its appearance *abstract*. Similarly, as demonstrated in [60], plausibility illusion can be affected if the laws of physics are challenged in a scene. In behavior manipulation, we can make the behavior of a scene naturally plausible in one scenario and implausible in another.

To satisfy the symmetry constraint, we will maintain a controlled physical environment to minimize interference from the physical world. This includes keeping the same room, lighting, seating arrangement, duration of the study session, breaks, order of the experiment among all users, and experiments.

To this end, we divide our experiment into two sets. In the first set, we vary the sense of presence by manipulating the *appearance* of the virtual object (place illusion). In the second set, we vary the sense of presence by manipulating the *behavior* of the virtual object (plausibility illusion). Each experiment will contain two blocks of trials: a *control* block and a *manipulated* block. In the *control* block, the scene contains virtual objects that have a natural appearance and plausible behavior. However, in the *manipulated* block, the scene contains virtual objects whose appearance is manipulated to be visually unnatural (unrealistic) or in which the virtual objects exhibit implausible behavior.

3.3 DC3 - Establishing a baseline

Since the presence itself has a subjective nature, it is logical that we also establish our baseline with a subjective measure. We establish the baseline using a subjective questionnaire, as its use is an accepted standard measure of presence in the research community [60], [85], [86]. Subjective questionnaires have been the standard measure of presence for

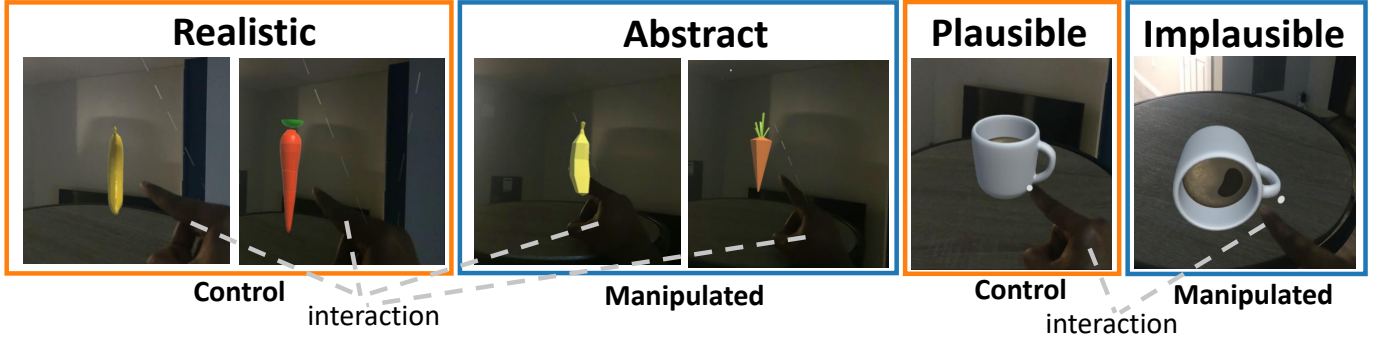


Fig. 2: Realistic, abstract, plausible, and implausible virtual objects used in the experiments.

many years and have been shown to be sensitive enough to find differences in presence when used to examine the difference between two visually similar fidelities [6]. The post-experience questionnaire provides scores that reflect the level of perceived presence in the scene. We measure various feelings of presence with the three most-cited and widely used questionnaires that measure the main aspects of presence. We use the Igroup Presence Questionnaire (IPQ) [10], [11], Slater-Usch-Steed (SUS) [5] and Witmer & Singer (PQ) [4].

3.4 DC4 - Selecting an interaction mechanism

We need a task that engages the participant and helps us measure our output variable, *reaction time*. Through interactions, the participant can physically be involved in the scene. In designing the interaction, we want to avoid imposing differential barriers to task completion between participants. An example of such effort would be to place a push button at an accessible height for all the participants.

Similarly, we want to avoid designing unnecessarily complex interactions that could potentially cause task performance to vary in unpredictable ways, independent of the targeted manipulations. Additionally, as part of our design, we ensure that the reaction time is not affected by external interventions. Therefore, we need to leverage existing elements in the taxonomy of the scene (cue, interactions, and feedback).

For this reason, we have employed a simple ‘tap’ as our interaction mechanism. It is precise and does not respond to direction and / or length of movement or any other physical attribute of a typical participant.

We designed a task that supports participants in knowing what to do (cue), knowing that the system is working (interaction/tap), and knowing if their action was understood by the system (feedback). In the *realistic vs. abstract* scenario, the appearance of the object in the scene is a cue for the user to take an action (tap) and the object leaves the scene as a feedback for a successful tap. In *plausible vs. implausible*, the change in coffee level is the cue to take an action (tap) and the cup changes color as a feedback for successful tap (more details in Section 4.6).

Next, we formally define the reaction time ($reaction_i$) in our study ($reaction_i$) for the trial i as follows:

$$reaction_i = interaction_i - cue_i$$

where trial i refers to one task iteration by the participant (appearance, tap, feedback). $interaction_i$ is the time when the tap is recorded in the trial i and cue_i is the time of onset of the task cue in the trial i .

4 USER STUDY

In this section, we detail the study measures, participants, equipment, and procedure for the user study.

4.1 Participants

We recruited 40 participants (23 male identifying, 16 female identifying, 1 non-binary identifying) with a mean age of 26.6 years (standard deviation of 5.5 years). All participants volunteered and provided an in-writing informed consent. They received \$25 for their participation. All but one participant had a technical background in computer science or engineering. All the participants had normal or corrected normal vision with contact lenses or glasses. 17.5% of the participants had 1 to 4 days per week MR experience, 24 participants had less than 1 hour per week experience, and 9 had never experienced MR before. Only 12 participants had used HoloLens 2 before the study. The study was granted ethics clearance according to the ethics and privacy regulations of our Institutional Review Board (IRB).

4.2 Material

The virtual environment for the study was presented on an ergonomic, untethered, self-contained holographic device, HoloLens 2 [26]. It has a second-generation custom-built Holographic Processing Unit (HPU) which enables real-time computer vision with enterprise-ready applications powered by a Qualcomm Snapdragon 850 CPU that runs all applications. All virtual scene applications were developed using the Unity 3D game development engine (2020.3.26f1) with API for the Universal Windows Platform on Windows 10 PC configured with SDK 10.0.19362.0.

HoloLens 2 is configured with the eye, spatial, and hand tracking input systems, along with the camera and other sensors with 43° horizontal 29° vertical 52° diagonal field of view. HoloLens 2 has two optical see-through displays [87] with a screen resolution of 1440 × 936 pixels (one for each eye) and a refresh rate of 60Hz. It uses a tinted visor to reduce the amount of light from the environment that reaches the user’s eyes.

4.3 Variables

In our study design we change two variables to test our hypothesis but we only manipulate one variable at a time. We chose *appearance* and *behavior* of virtual objects as our variables in the first and second experiment sets, respectively.

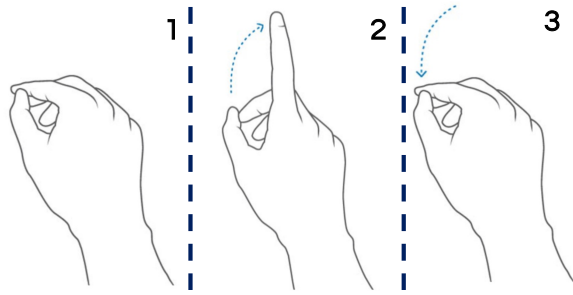


Fig. 3: Illustration of the experimental task using air tap.

Experiment Set 1: Realistic vs. Abstract

In the *control* trials, all virtual objects in the scene will have a realistic appearance (textured and natural) and plausible behavior. We refer to this block of trials as *realistic* in the rest of the paper. In the *manipulated* trials, all of the virtual objects will depict plausible natural behavior, but the appearance would be abstract (untextured and geometric). We refer to this block of trials as *abstract* in the remainder of the paper.

For *realistic* versus *abstract*, we need a scene that has textured virtual objects. To satisfy the simplicity constraint, we have modified the popular Fruit Ninja game [88] (1+ billion downloads and 2 million active players worldwide daily). To mimic the semantics of a multi-object environment, we needed at least two different objects that are similar in shape and size and different in textural properties. As we want a simple scene and one virtual object in a scene at a time, virtual objects will appear in the scene one after the other. In the end, we picked fruit (banana) and vegetable (carrot) for this experiment. We also removed complex interactions (slicing) and cognitive loads (scores) from the original game. In the *realistic* scene, we made the carrots and bananas appear as natural as possible in terms of their color, texture, and shape. In the *abstract* scene, we rendered the carrots and bananas with muted colors, without texture, and geometrically blocky shapes, as shown in Figure 2.

Experiment Set 2: Plausible vs. Implausible

In the *control* condition, the virtual objects in the scene have a natural appearance and plausible behavior. We will refer to this block of trials as *plausible* in the remainder of the paper. In *manipulated* condition, the virtual objects retain a natural appearance, but behave implausibly from real-world standards (disobey laws of physics). We will refer to this block of trials as *implausible* in the remainder of the paper.

For the *plausible* vs. *implausible* scenario, we chose a scene with elements that have physical constraints. We modified the popular Coffee Stack game [89] with a coffee mug, on a surface that periodically fills up with coffee. Like Fruit Ninja, we have also removed complex interactions and cognitive loads from this game. In the *plausible* scenario, the empty white porcelain coffee mug will be upright and the coffee will continue to pour into the mug periodically until the mug is full. Then, a new mug will appear, and this cycle will continue to repeat. In the *implausible* scenario, the empty white porcelain coffee mug is in a tilted position on the surface, and coffee continues to pour into and fill the mug even though gravity and volumetric constraints should not allow this (see Figure 2).

4.4 Experimental Task

The experimental task in both experiment sets can be divided into three sub-tasks, as shown in Figure 3. As a prerequisite, the objects in the experiments are placed in participant's field of view such that they can interact with the objects without lifting their arms, as illustrated in Figure 3(1) and captured in Figure 2. The second step for the participants is to lift their index finger, upon cue, to tap the virtual objects in the air, as shown in Figure 3(2). Finally, participants bring their finger back to the normal position as shown in the first step.

4.5 Measures

Presence scores for the questionnaires were obtained using a total of 39 items (6 SUS, 14 IPQ, 19 WS) on a 7-point scale. We did not modify any of the questions. The reaction time is recorded by our software on HoloLens 2. We collected 50 reaction time measurements per block of trials on average, totaling 200 data points per participant for the four blocks. The reaction time is measured in milliseconds (*ms*).

4.6 Pilot Study

Before starting the formal study, we conducted a pilot study with two participants. to tune the parameters to minimize environmental variables and maximize reliability and objectivity. We use a talk-aloud protocol and ask participants questions to align the experiment for general comfort and ease for the participants, but not to bias the experiment for a specific set of users. For session-specific parameters, we tuned the following parameters.

Participant position. We experimented with standing, walking, and sitting positions. The participants reported feeling tired while standing. While walking, they reported that the scene kept changing around them, necessitating extra attention to locate the virtual object. Participants reported the sitting position as the most comfortable position for the experiment. We asked the participant to sit on a chair with relaxed shoulders, an arm on the lap or armrest, and feet flat on the floor. The chair with armrest and backrest was reported to be the most comfortable with air-tap interaction [90].

Room lighting. Lighting affects a user's ability to see the virtual environment and its objects [91]. It also affects the rendering of virtual objects and user interaction with those objects, as they rely on the tracking module of HoloLens 2. For optimal visuals through HoloLens 2, lighting should be even and sufficiently bright so that a participant can see without effort, but not so bright so that a participant has a hard time looking into the environment. To compensate for the loss of the visor dim lights reflecting in the direction of the participant's head are deemed most effective because a tinted visor may cause a loss of contrast in the physical environment.

Experiment duration. How long will an experiment run with the same repetitive task? It should be long enough for the participant to feel involved and we can collect a sufficient amount of data on *reaction time*. It should be short enough that the participant does not feel tired or disengaged impacting the accuracy of their interactions. We tested a duration of 2 to 10 minutes. At 2 minutes, the participants

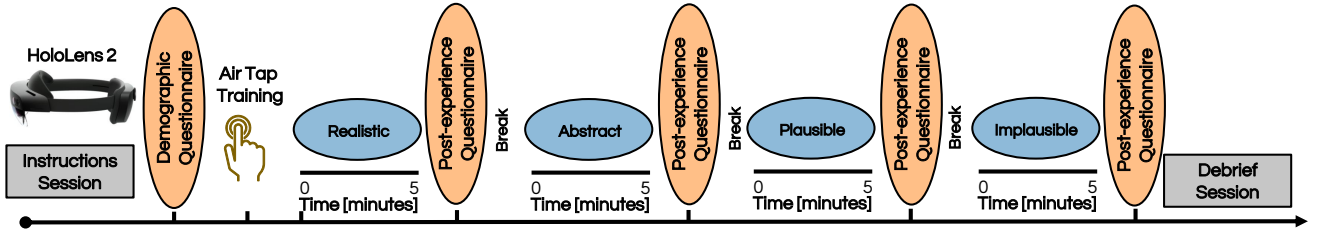


Fig. 4: User study timeline consisting of pre-and post-questionnaire as well as four blocks across the two experiments.

reported that they were unable to get acquainted with the environment. At 10 minutes, the participants reported feeling disinterested after a while. We also wanted to test the recovery time of the presence or reaction time (discussed in detail in 5.1.2). We picked 5 minutes per experiment as it allowed us to obtain at least 60 reaction time readings per experiment and the participant did not feel tired.

Wait time between blocks and experiments. After each block of an experiment, the participant was asked to complete the questionnaires. We explored 0-20 minutes of wait times after filling out the questionnaire and before starting the next block. Participants reported 5-minute wait times as sufficient, but a longer break was available upon request. In the main study, we explicitly asked participants before each experiment if they needed more downtime.

Experiment and block order. For an experiment, we could expose the participant to a control block and then to manipulated trials, or vice versa. In the latter case, there is a risk of obtaining better performance for the control block as a result of long exposure. In the former case, the user will get familiar with the experiment and its performance should improve for the manipulated block (a bad result for us). In our pilot study, we ran the first order with one participant and the second order with the other participant. We did not observe any difference between the two participants and decided to run all participants with control block first and manipulated block second. We agree that potential order effects warrant further investigation in the future, but we note that the ordering we chose is probably least likely to disadvantage task performance in the manipulated trials, due to the longer duration of prior practice.

Virtual object placement. We use Fitts's law [92] to calculate the expected time of motor movement for several different positions in the scene. We decided to place the virtual object such that it was parallel to the eye level of a sitting participant. The object was placed at a 45 centimeter distance, as recommended by HoloLens 2 intractable object guidelines with air tap interaction [93]. The size of the object was tested in the range between $1.4 \times 1.4\text{cm}$ and $3.5 \times 3.5\text{cm}$. Both participants felt comfortable with $1.4 \times 1.4\text{cm}$.

Event and task period. Each block in our experiments includes repetitive trials. If events are repeated too fast, task difficulty would increase, and task accuracy would suffer. If the events are far apart and the downtime between events could break the presence [60] We tested 1 to 15 second event periods; 3s periods were too small as participants reported being puzzled and unable to complete the task, participants felt bored for 8s periods. We chose 5 seconds as the period between events and tasks to keep the user comfortable while maintaining high task accuracy.

Cue appearance. We first tested the change in color as a

visual cue for the participant in both sets of experiments (see details in Section 3.2). We found that the color change as a cue was distracting. After trying additional cues, we used objects appearing in the scene as a cue for the realistic vs. abstract. For the coffee mug experiment, we tested the glowing button as a cue for tasks, but the participants found it distracting. At the end, we used filling the coffee in the mug as a cue. The participant found that it was more engaging than the color change and the glowing button.

Proximity of interaction. We tested the air tap interaction at a close and far distance. The participant struggled with the far air tap. The reason for this could be that the virtual object was placed near the user and the far-touch interaction moved the virtual object farther from the participant and created unnecessary distraction. The participant found the near air tap interaction is more natural, so we kept it as the mode of interaction for all of our experiments.

Feedback. For realistic vs. abstract, we chose virtual objects disappearing as a feedback for successful interaction, just like fruit ninja game. Participants found it acceptable. For plausible vs. implausible experiments, feedback was tricky without adding another virtual object in the environment. We tested with a subtle coffee-poured sound as feedback. One of the participants found it distracting. We then decided to use the mug color change as feedback rather than a cue and tested again. Both participants found that the color change feedback was more subtle and less distracting. Interestingly, the same "change in color effect" participant found it distracting when putting it as a cue instead of feedback.

4.7 Procedure

Figure 4 shows the complete timeline of the user study. After arriving at the study room, the experimenter welcomed the participants and gave them a verbal introduction and instructions. Participants then read and signed the IRB-approved informed consent form and filled out a pre-questionnaire about demographic characteristics (age, self-identified sex, and familiarity with the Mixed Reality (MR)). We also briefed them on MR and HoloLens 2). We then explain the flow of the study, the types of experiments, the duration of each block and experiment, their order, and the questionnaires. We further explained how to interact with the virtual objects, how to perform an air tap, and how hands should be held in their laps when not tapping. We also explain what to expect while wearing the headset, how and when they should interact with the virtual object, and what the cue would be for tapping and feedback.

As speeding could potentially override participants' natural instincts, we advised them on the importance of completing the task correctly, without the need to rush or stall. To avoid this instinctual bias, we deliberately kept this information (reaction time measurement) from the participant at

No	realistic (μ , MAD)	abstract (μ , MAD)	F	p (<0.05)
ALL	4.97, 0.90	2.74, 0.74	107.63	yes
SUS	4.92, 1.09	2.60, 1.89	70.89	yes
IPQ	4.17, 1.58	2.31, 0.57	38.99	yes
PQ	5.58, 0.65	3.09, 0.99	122.11	yes

TABLE 1: Mean, mean absolute deviation (MAD), and F-test results for all questionnaires.

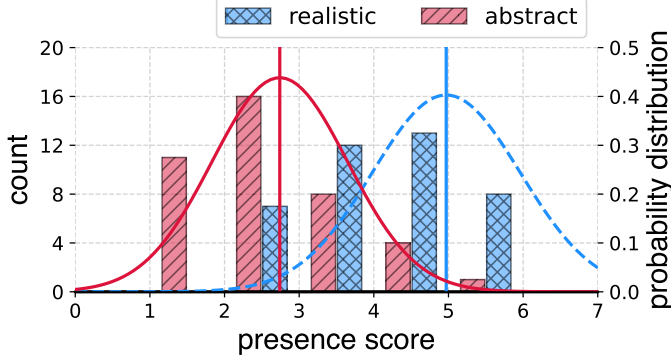


Fig. 5: Histogram, means (left y-axis), and fitted Gaussian distribution (right y-axis) using mean questionnaire scores.

the beginning and added this information in the debrief. Participants were also not informed of the frequency of tasks to minimize anticipation. We also used the talk-aloud protocol and asked participants to think aloud as they performed the tasks to assess the varied reactions of the participants overtime, which the subjective questionnaire does not capture. We also monitor the entire process through the first-person user view in Mixed Reality Capture [94]. Participants were also advised to stop at any point in the experiment if they felt uncomfortable.

We world-locked the virtual scene in the environment and put the headset on the participant. The reason to world-lock the scene is to remove the environment variable, and world-lock is adjusted at participants' eye level by adjusting the chair of the participant. We adjusted the headset to the participant's head and calibrated it to participant's interpupillary distance for the best visual results. Before the first experiment, for training purposes, participants were then asked to interact with the virtual object (an empty dialog box with the next button and cancel) using an air tap. The feedback was set to sound that both the experimenter and the participant could hear when the first air tap was successful.

We set the order of blocks for the participant as *realistic*, *abstract*, *plausible*, and finally *implausible*. In each block, the timer is started only when the participant is in a comfortable position and verbally confirms readiness. At the end of the 5 minute period, the block ends by closing the scene (through the device portal [95]). We then remove the headset from the participant's head and ask them to fill out three post-experience questionnaires (see more details in Section 3.3). After completing four blocks and post-experience questionnaires, we thanked the participants for their participation and provided their compensation. The participants were then briefed on the study. After informing the participants about the purpose of the study, at this stage we provided them with the option to withdraw from the study.

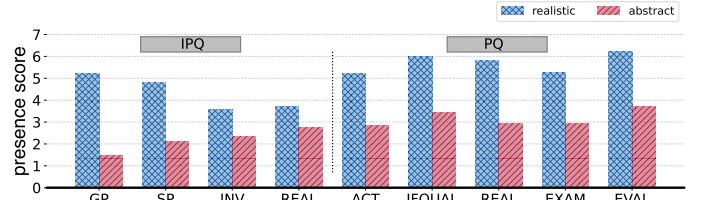


Fig. 6: Questionnaire Subscale Scores. IPQ has general presence (GP), spatial presence (SP), involvement (INV), and realism (REAL); PQ has possibility to act (ACT), interface quality (IFQUAL), realism (REAL), possibility to examine (EXAM), and self-evaluation of performance (EVAL).

4.8 Hypothesis

Based on the related work, we developed the following hypothesis for our experiment sets.

- H1: Manipulating the place illusion (appearance of virtual object) or plausibility illusion (behavior of virtual object) leads to change in the presence.
- H2: Change in presence score is due to the virtual object manipulations and could not have happened by chance.
- H3: Change in presence for a participant leads to change in participant's reaction time.
- H4: Change in reaction time is due to the change in presence and could not have happened by chance.
- H5: Presence and reaction time are negatively correlated.

5 RESULTS

In this section, we present the results quantifying *presence* using questionnaires and user *reaction time*.

5.1 Experiment Set 1: Realistic vs. Abstract

In this experiment, we changed the feelings of *presence* by manipulating the appearance of the object (i.e. *place illusion*). We get presence scores from the questionnaires and reaction time scores from HoloLens 2.

5.1.1 Presence Questionnaire Scores

We collect answers to all questionnaires on a 7-point scale. We obtain the *presence* score by averaging the 7-point scores of all the questionnaires individually and combined. We use non-parametric analysis of variance (ANOVA) tests to compute statistical results, since all questionnaires use an ordinal scale, which means that the answers are in order, but without a degree of difference between choices. We use Python's Scipy library to do all the calculations [96].

Questionnaire scores. The results for this experiment are reported in Table 1. We found that manipulating the object appearance had a significant impact on *presence*. We observed $F(1, 38) = 107.63, p < 0.001$ for all the questions, $F(1, 38) = 70.89, p < 0.001$ for SUS, $F(1, 38) = 38.99, p < 0.001$ for IPQ, and $F(1, 38) = 122.11, p < 0.001$ and PQ questionnaires¹. Figure 5 shows the histogram of the scores across users and its probability distribution.

Score variance. We take mean absolute difference (MAD) from the mean as a measure of variance. Table 1 presents MAD values across all the questionnaires. We use F-test to

1. $F(X, Y)$: X, Y are degrees of freedom between and within groups, respectively. X = total groups-1, Y = group size-total groups.

Experiment	Reaction Time (ms) (μ — MAD)	Change (%)	No. of Interactions
realistic	954 — 170		50
abstract	1313 — 290	37.63	44

TABLE 2: Average, mean absolute difference (MAD), %age change of user reaction times, and average number of interactions across the two sets of experiments.

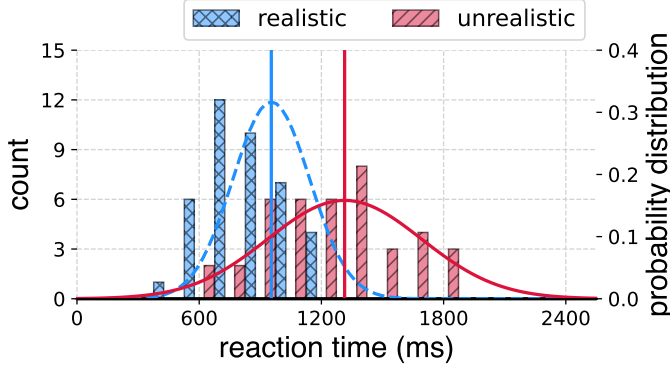


Fig. 7: Histogram, means (left y-axis), and fitted Gaussian distribution (right y-axis) using the average reaction time across participants.

check if the variance measures across the blocks are equal. Our F-test scores are $F(1, 38) = 0.980, p = 0.11$, suggesting that the variance of the scores did not change across blocks.

Subscales. The mean scores for all subscales are shown in Figure 6. Across IPQ subscales, general presence (*GP*) and spatial presence (*SP*) are the two most significantly changed aspects of the questionnaire. The user involvement (*INV*) and realism (*REAL*) also changed, but not to the same degree. Across all subscales, we observed at least 1-point difference in scores on a 7-point score scale. Across the PQ subscales, no single subscale had a significantly higher change than the other subscales.

Additional statistical analyses. We measured the correlation coefficient between overall scores across all questionnaires before and after manipulation. We found that the scores are correlated with the Pearson correlation coefficient of 0.53 for the appearance manipulation experiment. We conducted paired t-tests for the two blocks in the experiment and found a t-statistics value of 19.44 and a p-value of less than 0.001. A similar analysis for the IPQ questionnaire suggested an even higher correlation value of 0.67. PQ questionnaire showed similar results (omitted for brevity).

User characteristic and presence. We performed a regression analysis using F-test to check if age, gender, and familiarity with the MR had any impact on the *presence* scores. In this analysis, the null hypothesis is that a regression model based on a given variable is not a better fit than a simple intercept-only model. We observed that age, gender, and familiarity with MR did not have any effect on overall *presence* scores, as we obtained values of $F(1, 38) = 0.9561, p = 0.3412$, $F(1, 38) = 1.6370, p = 0.96$, and $F(1, 38) = 2.192, p = 0.7132$, respectively.

5.1.2 Reaction Time

To evaluate the *reaction time* of participants, we collect timestamps when user **interacts** in response to *cue* and use the difference between two timestamps as *reaction time*. Since our experiments run for at least 5 minutes, we get around

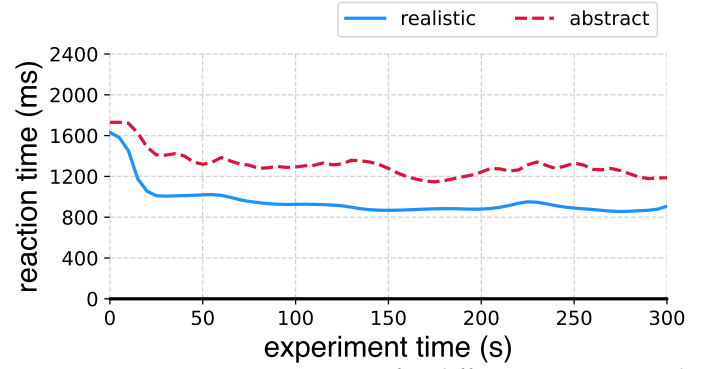


Fig. 8: Average user *reaction time* for different experimental settings. User reaction time recovers (decreases) over time.

50 data points per user per experiment. We use Python's Scipy library to do all the calculations [96]. Table 2 presents the high-level results of the experiments.

Reaction time scores. Our results show that average user reaction times increased from 954 ms to 1313 ms for the object *appearance* manipulation experiment. This represents a significant change of 37.63% for appearance manipulations in the user reaction time across experiments. Furthermore, we also measured the number of times a user was able to respond to the cue. Since the number of cues between experiments differed due to slight variations in the experiment duration, we only considered the first 60 cues to collect these statistics. We see that the number of interactions within a given set of experiments decreased after the manipulations. **Reaction time variance.** Figure 7 shows the distribution of average reaction time scores across the users. We analyze how mean and variance compare between experiments. The F-test scores, $F(1, 38) = 35.74, p < 0.001$, for the mean user reaction time between realistic and abstract experiments suggested that both samples are equally distributed from their means. We also performed the variance test across experiments using the absolute difference of the mean as a measure of variance. The results showed that the variance measure did not change, as we obtained $F(1, 38) = 12.96, p = 0.0005$ for the experiment.

Reaction time recovery. We next looked at the reaction time of users over time. In Figure 8, we see that participants took significantly longer to respond to the cues at the start of the experiment. The reaction time drastically dropped and settled to a steady-state within the first 30 seconds for the realistic vs. abstract experiment (a). After that, there was only a modest improvement in the reaction time over the duration of the experiment.

5.1.3 Experiment Set 1: Discussion

Our results suggest that our first hypothesis “H1: *manipulating the place illusion (appearance of virtual object) leads to change in presence*” is valid. We successfully altered the presence by simply manipulating the appearance of the object from realistic to abstract. Prior work on manipulating the place illusion to alter the presence also supports our results [60]. Our second hypothesis “H2: *change in questionnaire scores was due to manipulations in place illusion*” can be accepted, as the questionnaire scores before and after our manipulations are spread equally from their respective means. This is a standard test to measure if there are external factors

No	plausible (μ , MAD)	implausible (μ , MAD)	F	p (<0.05)
ALL	5.17, 0.93	2.81, 1.091	92.55	yes
SUS	5.34, 1.09	2.70, 1.13	72.79	yes
IPQ	4.51, 1.35	2.63, 0.90	38.56	yes
PQ	5.61, 0.79	2.98, 1.30	86.51	yes

TABLE 3: Mean, mean absolute deviation (MAD), and F-test results for all questionnaires.

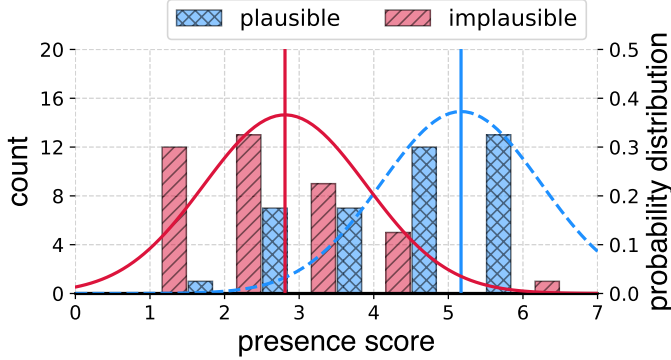


Fig. 9: Histogram, means (left y-axis), and fitted Gaussian distribution (right y-axis) using mean questionnaire scores.

that contributed to change in the repeated measure factor. Additionally, high correlation between questionnaire scores before and after the manipulation suggests that the users who felt a greater presence before the manipulation were also the ones who felt a low presence after the manipulation.

Our third hypothesis “H3: change in presence leads to change in user reaction time” is also valid. Our first hypothesis confirms that the presence changed from the experiment with a realistic object to the experiment with an abstract object. Simultaneously, we see a significant increase in the user reaction time as participants moved from realistic to abstract object experiments. Looking at the results, we can accept our fourth hypothesis “H4: change in reaction time was due to change in presence”. Just like our second hypothesis, we use ANOVA test to establish that the change in reaction time was due to change in presence.

Finally, the recovery time analysis also yields interesting points. The reaction time for the manipulated experiment block is always higher than the control experiment block. Furthermore, the reaction time reaches a steady-state after an initial few seconds. This means that the participants initially took some time to get acquainted with the environment, which increased their reaction time. As time went on, they interacted with the virtual objects in a fairly consistent manner.

5.2 Experiment Set 2: Plausible vs. Implausible

As in the previous experiment set, we get presence scores from questionnaires and reaction times from HoloLens 2.

5.2.1 Presence Questionnaire Scores

In this experiment, we changed the feelings of presence by manipulating the behavior of the object (i.e. *plausibility illusion*). The questionnaires used, analysis setup, and participants are the same as the previous experiment set.

Questionnaire scores. Table 3 reports the results for this experiment. We found that altering the object behavior also

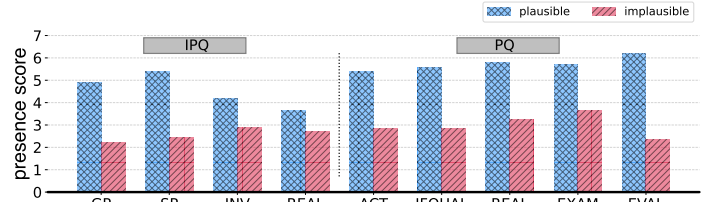


Fig. 10: Questionnaire Subscale Scores. See Figure 3 caption for subscale acronyms.

Experiment	Reaction Time (ms) (μ — MAD)	Change (%)	No. of Interactions
plausible	930 — 240		54
implausible	1182 — 210	27.10	51

TABLE 4: Average, mean absolute difference (MAD), %age change of user reaction times, and average number of interactions across the two sets of experiments.

had a significant effect on the overall *presence*. We observed $F(1, 38) = 92.55, p < 0.001$, $(F(1, 38) = 72.79, p < 0.001)$, $(F(1, 38) = 38.56, p < 0.001)$, and $(F(1, 38) = 86.51, p < 0.001)$, for all questionnaires, SUS, IPQ, and PQ, respectively. Figure 9 shows the histogram of the scores across users and its probability distribution.

Score variance. We take mean absolute difference (MAD) from the mean as a measure of variance. Table 3 presents MAD values across all the questionnaires. We use F-test to check if the variance measures across the blocks are equal. The F-test scores are $F(1, 38) = 1.250, p = 0.64$, suggesting that variance of the scores did not change across blocks.

Subscales. The mean scores for all subscales are shown in Figure 10. As in experiment set 1, general presence (GP) and spatial presence (SP) significantly changed and we observed at least 1-point difference in scores on a 7-point score scale. For PQ, change across subscales is consistent.

Additional statistical analyses. For the behavior manipulation experiment, we found a Pearson correlation coefficient of 0.48, t-statistics value of 11.74, and a p-value of less than 0.001 across blocks in the experiment. Individual questionnaires also showed higher correlation such as 0.57 for IPQ, but additional results are omitted for brevity.

5.2.2 Reaction Time

We collected 50 data points per user per experiment on average. Table 4 presents the results for the experiments.

Reaction time scores. Our results show that average user reaction times increased from 930 ms to 1182 ms for the object *behavior* manipulation experiment, representing a significant change of 27.10% across experiments. As in the first experiment set, we see a decrease in the number of interactions after the manipulation.

Reaction time variance. Figure 11 shows the distribution of average reaction time scores across the users. The F-score for plausible vs. implausible experiment was smaller than the first experiment set, $F(1, 38) = 9.77, p = 0.002$, but statistically significant enough to prove that participants showed very similar behavior in terms of their distribution of means. Our results showed that the variance measure did not change, as we obtained $F(1, 38) = 7.04, p = 0.0009$ for the experiment.

Reaction time recovery. Figure 12 shows the reaction time of users over time. The participants took slightly longer

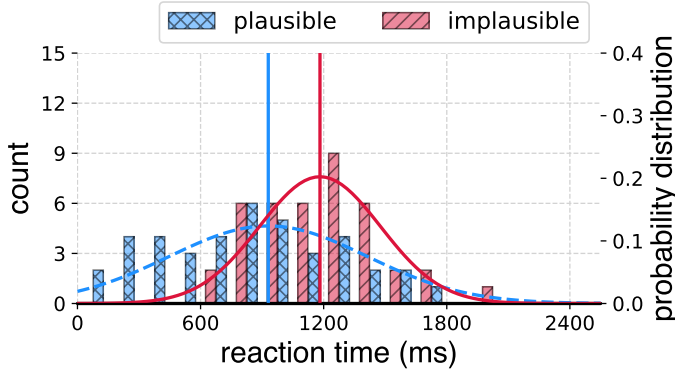


Fig. 11: Histogram, means (left y-axis), and fitted Gaussian distribution (right y-axis) using the average reaction time across participants.

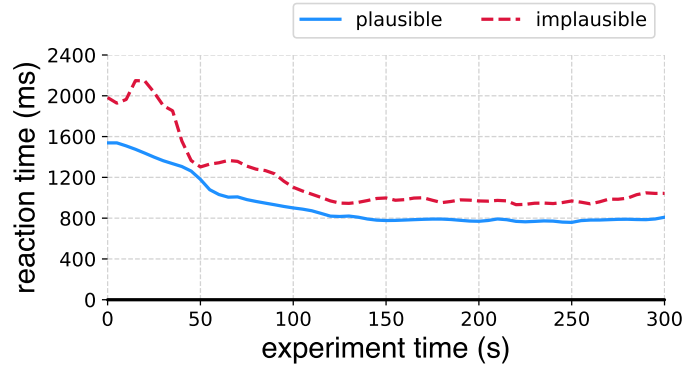


Fig. 12: Average user reaction time for different experimental settings. User reaction time recovers (decreases) over time.

than 60 seconds to reach a steady-state for the plausible vs. implausible experiment, longer than the first experiment set.

5.2.3 Experiment Set 2: Discussion

The purpose of this experiment was to establish that presence can be modified in ways other than manipulations to the place illusion done in the previous experiment. For this experiment, our hypothesis “H1: manipulating the plausibility illusion (behavior of virtual object) leads to change in presence” is valid. We altered the presence by simply manipulating the behavior of the object from plausible to implausible. Gravity is a very important aspect of our life and we expect objects to behave in specific ways under gravity. If their behavior is not plausible according to the laws of physics, it leads to degraded presence. As with our place illusion experiment, our distribution of variance analysis suggests that the second hypothesis “H2: change in questionnaire scores was due to manipulations in plausibility illusion” can be accepted. Similar to the previous experiment, we saw a high correlation between questionnaire scores suggesting that plausibility illusion was the factor in altering presence.

The remaining hypotheses, H3-H4, can be accepted based on our results. The only difference is that the change in presence was due to the change in plausibility illusion and not the place illusion. This also demonstrates that reaction time changes irrespective of how the presence was changed. This, combined with our modest correlation scores, means that other factors may be affecting the presence other than our place and plausibility illusions. The recovery time analysis for this experiment set also yields similar results as the previous experiment. However, the users take longer time to reach the study state and there are significant variations in the implausible experiment. This means that the manipulating the appearance of the objects has a less effect on participants than the behavior manipulation. This is understandable as humans are more likely to see objects with non-standard appearance as opposed to experiencing objects that do not conform to gravity.

6 DISCUSSION

In this section, we discuss the implications of the quantitative results presented in the previous section.

Our first hypothesis “H1: Manipulating the place illusion (appearance of virtual object) leads to change in the presence”

for the first experiment set can be accepted. The presence score of participants for the manipulated experiment was more than 2 points lower, on a 7-point scale, than the control experiments. As we used everyday virtual objects, bananas in this case, a change in its texture significantly alters our strong experience-based prior. The results from the second experiment set further supports this statement and results in acceptance of H1. As all the participants experience gravity at all times, an object defying gravity is implausible and causes a break in presence. This statement is supported by prior work asserting that break in plausibility, gravity defying behavior, leads to change in presence [60].

In our individual experiment set discussion, we have stated that the second hypothesis “H2: Change in presence score is due to the virtual object manipulations and could not have happened by chance.” is valid as variance measure does not change across the two blocks within a given experiment set. This further suggests that the questionnaire measures were a reliable measure for quantifying presence in our experimental setup. This is an important observation that supports our second contribution, which establishes that the questionnaires used for the fully virtual environments can also be used in the mixed reality environments. Finally, our two hypotheses are acceptable for the individual questionnaires and their subscales, as change in presence and variance measure behavior is the same as the overall results.

Our user reaction time results and their variance measures suggests that our third “H3: Change in presence for a participant leads to change in participant’s reaction time” and fourth “H4: Change in reaction time is due to the change in presence and could not have happened by chance.” hypothesis can be accepted as well. Furthermore, as the users became more familiar with the environment and setup, the reaction times from one set of experiments to the other experiment decreased. We also see an improvement in the number of interactions across the two sets. However, this was not able to overcome the increase in reaction times due to our manipulations within a set of experiments. Within a block, across both experiments, the reaction time drastically improved at the start but quickly reached a steady-state. The difference in reaction time between the blocks of an experiment remained constant over time. This indicates that the recovery effect is consistent across blocks and our manipulations influenced the change in reaction times of the participants.

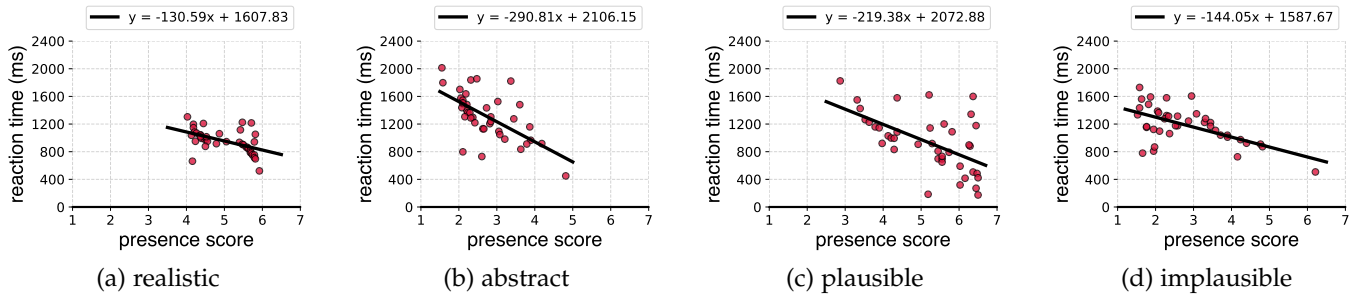


Fig. 13: Reaction Time vs. Presence: Reaction time decreases as the presence increases. Reaction time and presence also show modest correlation: overall (-0.65), realistic (-0.51), abstract (-0.63), plausible (-0.57), and implausible (-0.59). Each red circle represents a study participant. Black line is the linear regression fit for the data.

6.1 Presence - Reaction Time Correlation

Till this point, we have established that we were able to change the presence and participant's reaction times changed as presence changed. Figure 13 shows the scatter plots of participants' *presence* scores and their *reaction times* for all the experiments. We observe that *reaction time* has an inverse correlation with *presence* score, albeit with different slopes for the linear regression fitted model across experiments. Our results highlight that appearance and behavior manipulations impacted the presence questionnaire scores and reaction time values in the same manner when aggregated across all users. We also observed a moderate correlation between presence scores and reaction times of individual users. This indicates that a user that experienced a high presence score in the control experiments also had a small reaction time in the control experiments.

This is a significant result and suggests that *presence* impacts *reaction time*, at least in the confines of our study. This means that we can accept our hypothesis that "*H5: presence and reaction times are correlated*". However, the correlation values lie in the modest range between -0.5 to -0.65. This means that further analysis is needed to fully establish that reaction time can be used to quantify presence. This is because a given change in *presence* score or *reaction time* does not elicit the exact same change in the other variable across all experiments. Also, the individual differences between different experiments are significant enough to warrant that the use of this relationship should be scenario specific. Having said that, we can at least state that if the presence of a user increases or decreases, the user reaction time will also decrease or increase, respectively.

7 LIMITATIONS AND FUTURE WORK

For this study, we used two scenes consisting of a simple one-object scenario, a design choice to minimize the effect of variables other than presence on the reaction time. Therefore, the setup may not be sensitive to the broader effects of varying the feelings of presence on reaction time in a multi-object virtual scene. Future work should investigate how presence relates to reaction time with different degrees of scene complexity, cognitive load, and dynamic physical environment. Similarly, we investigated the effects of varying the feelings of presence with only periodic tasks and active interaction (tap). It is worth investigating how the presence and reaction time can be affected in other types of non-periodic tasks and passive interaction (eye gaze). Further-

more, participants reported their presence levels using the questionnaires that leverage Likert scales. Other questionnaires that rely on different assessment mechanisms such as open-ended questions might reveal additional insights into the results. We also acknowledge that our proposed approach depends on user interactions to measure the reaction time. In virtual scenes with little or no interaction, our technique may fail to produce any measurement. In future work, the measurement of reaction time should consider using some other backup mechanism, such as eye-gaze tracking, to combat low-interaction scenarios.

We have not found any effect of gender, age, and familiarity with MR on the presence or reaction time in this study. This can be due to our purposefully simple design, but in a complex scenario with some applications, these user characteristics may have an effect on the presence or reaction time. In addition, we have not tested the effect of break in the presence, but it may have some impact on the presence and reaction time. In the future, the break in the presence can be tested with the reaction time as a measure and may add to the discussion of the presence-reaction-time relationship. In this study, we did not ask participants to fill out a cybersickness questionnaire. However, during the study with participants, we asked participants to report any discomfort they feel during or at the end of the experiments. None of the participants reported any feeling of discomfort. This may be due to the limited exposure time and a wider adjustable range of interpupillary distance that Hololens 2 offers [26]. However, in the future, this variable can be tested in isolation to isolate the potential impact of cybersickness in varying the feelings of presence.

Despite the limitations of the work and the opportunity for improvements, we argue that our results present sufficient evidence of a relationship between *presence* and *reaction time* to justify a further discussion of whether a performance-based metric such as *reaction time* can be used to describe *presence*. Currently, post-experience questionnaires are the most commonly used measures of *presence* in previous work. However, a major disadvantage of such questionnaires is that they are based on the subjects' memories of the experience. Such memories can reflect an inconsistent and incomplete picture of the experience. *Reaction time*, on the other hand, is a passive and objective measure that does not depend on the comprehension of the user and the memory of the experience. In the future, with additional investigation, it can serve as a measure of *presence* and as a feedback loop that developers can use to improve the run-

time experience, since it measures the phenomenon at the moment when it is perceived.

8 CONCLUSION

We presented a user study (N=40) to understand the relationship between *presence* and *reaction time*. We changed the sense of *presence* of the participants by manipulating *appearance (place illusion)* and the non-task-relevant *behavior (plausibility illusion)* of the virtual object, and systematically measured the *reaction time* of the participants in response to visual stimulus. Our post-experience questionnaires show a significant change in the *presence* across experiments. Similarly, we see a significant change in user *reaction time* as we vary feelings for *presence*. Our analysis shows a negative correlation between the *presence* and *reaction time*. Furthermore, our study provides insight into the considerations for using *reaction time* as a possible measure of *presence*, as well as preliminary recommendations on the possibilities of future research to better understand the relationship between *presence* and *reaction time*. We found that as the average *presence* score for the two illusions decreased from 4.97 to 2.74 and 5.17 to 2.81 (on a 7-point scale), the average *reaction time* increased by 37.63% and 27.10%, respectively. Although our study points to an inverse relationship between *presence* and *reaction time*, we also note that the slope of the negative correlation may depend on the scenario under consideration and needs further investigation.

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